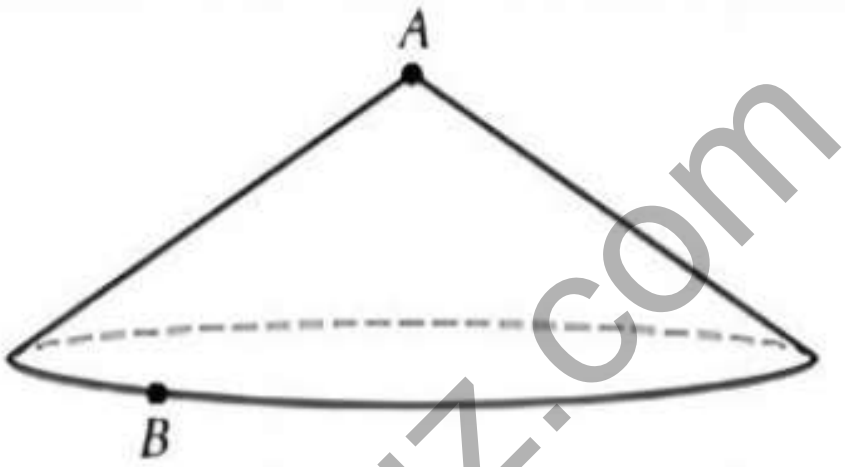
20 Mark for Review

Right rectangular prism X is similar to right rectangular prism Y. The surface area of right rectangular prism X is **59 square centimeters** $(\mathbf{cm^2})$, and the surface area of right rectangular prism Y is $\mathbf{1{,}475\ cm^2}$. The volume of right rectangular prism Y is **1,500 cubic centimeters** $(\mathbf{cm^3})$. What is the sum of the volumes, **in** $\mathbf{cm^3}$, of right rectangular prism X and right rectangular prism Y?

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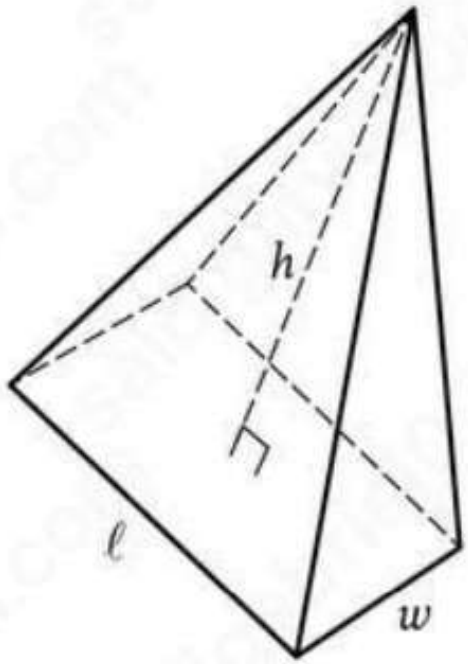
Note: Figure not drawn to scale.

For the right circular cone shown, B is a point on the circumference of the base, and t
length of segment AB (not shown) is 32 centimeters. If the height of the cone is 16
centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k

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Mark for Review



Note: Figure not drawn to scale.

The figure shown is a right rectangular pyramid, where $\ell = 16$ units, $w = 8$ units, and $h = 18$ units. What is the surface area, in square units, of the pyramid?

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Mark for Review

A right square pyramid has a height of 12 inches and a volume of 196 cubic inches. What is the perimeter, in inches, of the base of this pyramid?

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	Area (square inches)	Perimeter (inches)
Rectangle A	660	220
Rectangle B	2,640	n

What is the value of n?

A. 2,200

B. 1,760

C. 880

D. 440

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The length of a rectangle's diagonal is $\sqrt{65}$, and the length of one of the rectangle's sides is 4. What is the perimeter of the rectangle?

(A) 11

(B) 22

(C) 28

(D) 65

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14 Mark for Review

A rectangular poster has an area of 330 square inches. A copy of the poster is made in which the length and width of the original poster are each increased by 50%. What is the area of the copy, in square inches?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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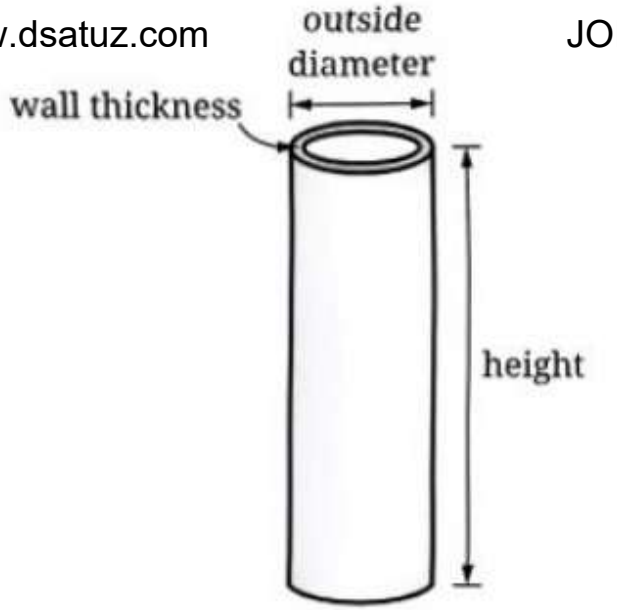
A right triangular pyramid has exactly six edges. Each of these edges is 96 centimeters long. If the surface area of the pyramid is $k\sqrt{3}$ square centimeters, what is the value of k?

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	Volume (cubic units)
Right circular cylinder A	32π
Right circular cylinder B	864π

The table shows the volume of two similar solids, right circular cylinder A and right circular cylinder B. The radius of right circular cylinder A is 2 units. The surface area of right circular cylinder A is $k\pi$ square units, and the surface area of right circular cylinder B is $n\pi$ square units, where k and n are constants. What is the value of $n - k$? (The surface area of a right circular cylinder with radius r and height h is $2\pi r^2 + 2\pi rh$.)

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Note: Figure not drawn to scale.

In the figure, a hollow steel pipe is in the shape of a right circular cylinder. The steel pipe has an outside diameter of 48 inches, a wall thickness of $\frac{3}{8}$ inches, and a height of 138 inches. Which of the following is closest to the volume, in cubic inches, of the wall of this steel pipe?

A. 61

B. 325

C. 1,242

D. 7,743

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15 Mark for Review

A rectangular region of land is divided into 88 square lots of equal area A, in square units. The length of the region is 5.5 times the width of the region. If the length of the region is $x\sqrt{A}$ units, what is the value of x?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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16 Mark for Review

A right square pyramid has a surface area of $100 + 20\sqrt{146}$ square inches, which includes a base area of 100 square inches. What is the height, in inches, of this pyramid?

Enter your answer (e.g., 5.566, -5.566, 2/3, -2/3)

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